Spatial Population Dynamics: Fertility, Migration, and Policies

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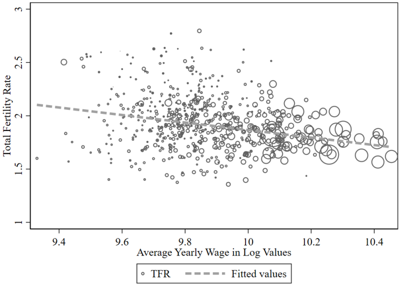
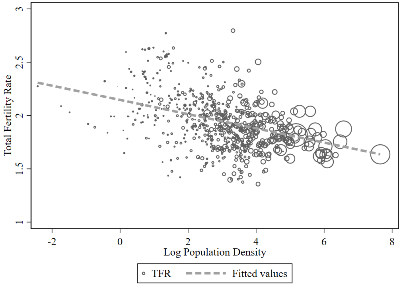
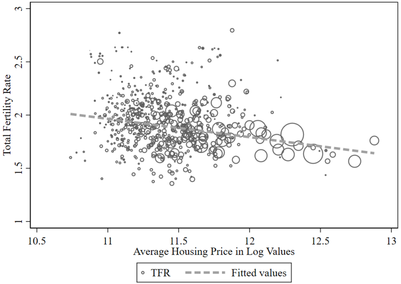
Motivation

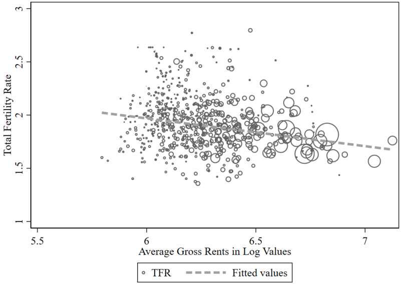
- Fertility has fallen sharply across advanced economies
  - $\rightarrow$ U.S. TFR: 2.1 (1990) $\rightarrow$ 1.6 (2023)
- Striking empirical regularity: lowest fertility in most productive cities
  $\Rightarrow$ Spatial reallocation toward productive cities may come at cost of lower aggregate fertility
- Existing explanations (gender roles, labor frictions) are often silent on spatial dimension
- **Key insight**: Fertility should be analyzed as equilibrium outcome of a spatial household problem

Motivating Facts









This Paper

1. Develop a quantitative spatial model integrating endogenous fertility, migration, and housing markets
2. Derive expressions on how local conditions affect fertility and propagate across space
3. Calibrate to 741 U.S. commuting zones and evaluate place-based policies
4. Characterize optimal interregional transfers under different welfare objectives

Key Findings

1. Housing and children are strong complements ($\rho = 0.49$)
   $\Longrightarrow$ High housing costs reduce both housing consumption and fertility
2. Housing supply reform raises aggregate population growth by 1 percentage point (0.92 $\rightarrow$ 0.93)
3. Optimal transfers depend on welfare objective:
   - $\rightarrow$ Maximizing per-capita welfare: concentrate workers in large CZs
   - $\rightarrow$ Maximizing total welfare: retain workers in smaller CZs
4. Endogenous fertility acts as stabilizing force in spatial dynamics

Literature

- Quantitative spatial economics (Redding & Rossi-Hansberg 2017, Allen & Arkolakis 2014)
- Economics of fertility (Becker 1960, Barro & Becker 1989)
- Housing and fertility (Dettling & Kearney 2014)
- Housing supply constraints (Hsieh & Moretti 2019)
- Concurrent work: Arkolakis et al. (2026) on South Korea

Roadmap

- Model
- Model mechanisms
- Calibration and model fit
- Counterfactual analysis
- Conclusion

Model

Environment

- N regions, discrete time, two-period OLG
- **Young**: reside with parent, no decisions
- **Adult**: observe location preference shocks, choose:
  - $\rightarrow$ Where to live (migration)
  - $\rightarrow$ Consumption, housing, fertility
- Each region produces differentiated variety (CES aggregation)
- Housing supplied with region-specific elasticity (Saiz 2010)

Equilibrium Definition

Preferences

Adult agent ν moving from o to i obtains tenure status $j \in \{O, R\}$ with probability ζ_i and derives utility:

$$U^j_{oi,t}(\nu) = \frac{B_{i,t}(\nu)}{d_{oi,t}} V^j_{i,t}(\nu)$$

Nested CES utility:

$$V^j_{i,t} = \left[\alpha^{\frac{1}{\gamma}} C^{\frac{\gamma-1}{\gamma}} + (1-\alpha)^{\frac{1}{\gamma}} \left[\beta^{\frac{1}{\rho}} h^{\frac{\rho-1}{\rho}} + (1-\beta)^{\frac{1}{\rho}} n^{\frac{\rho-1}{\rho}}\right]^{\frac{\rho}{\rho-1}\frac{\gamma-1}{\gamma}}\right]^{\frac{\gamma}{\gamma-1}}$$

- γ: elasticity between consumption and housing-fertility composite
- ρ: elasticity between housing and children
- $B_{i,t}(\nu)$: idiosyncratic taste shock (Fréchet distributed)

Budget Constraint and Production

Household budget constraint:

$$P_{i,t}C^j_{i,t} + r_{i,t}h^j_{i,t} = w_{i,t}(1 - \chi_{i,t}n^j_{i,t}) + T^j_{i,t}$$

- Children require $\chi_{i,t}$ units of parental time each
- Opportunity cost: $\chi_{i,t}w_{i,t}$ per child

Production:

$$\text{Unit cost: } uc_{i,t} = \frac{w_{i,t}}{A_{i,t}(L^e_{i,t})^\psi}$$

- $\psi \geq 0$: agglomeration externality
- $L^e_{i,t} = (1 - \chi_{i,t}n_{i,t})L_{i,t}$: effective labor

Housing Supply and Revenue Distribution

Housing supply (Saiz 2010):

$$r_{i,t} = \bar{r}_{i,t} H_{i,t}^{\xi_i}$$

- ξ_i: inverse housing supply elasticity (higher $\Rightarrow$ more inelastic)

Housing revenue distribution:

$$T_{i,t}^{O} = \frac{1}{\zeta_i} \left[\theta \cdot \frac{r_{i,t} H_{i,t}}{L_{i,t}} + (1-\theta) \cdot \frac{\sum_{k=1}^{N} r_{k,t} H_{k,t}}{\sum_{k=1}^{N} L_{k,t}} \right], \quad T_{i,t}^{R} = 0$$

- ζ_i: probability of being homeowner in region i
- Local revenue + national redistribution to homeowners

Optimal Fertility

Household optimization yields:

$$n_{i,t} = \frac{1}{\chi_{i,t} w_{i,t}} \cdot \frac{(1-\beta)(\chi_{i,t} w_{i,t})^{1-\rho}}{\beta r_{i,t}^{1-\rho} + (1-\beta)(\chi_{i,t} w_{i,t})^{1-\rho}} \cdot \frac{(1-\alpha)\Psi_i^{\frac{1-\gamma}{1-\rho}}}{\Xi_i} \cdot (w_{i,t} + T_{i,t})$$

Key economic forces:

- Fertility depends on ratio $r_{i,t}/(\chi_{i,t} w_{i,t})$
- When $\rho < 1$: housing and children are complements
- Higher wages: substitution effect $-$, income effect $+$
- Higher rents: price effect $-$, wealth effect $+$

Migration and Population Dynamics

Given Fréchet preferences, migration share from o to i:

$$\frac{L_{oi,t}}{n_{o,t-1}L_{o,t-1}} = \frac{B_{i,t}(d_{oi,t}^{-1}V_{i,t})^{\eta}}{\sum_{k=1}^{N} B_{k,t}(d_{ok,t}^{-1}V_{k,t})^{\eta}}$$

Key features:

- Fertility in numerator through origin's $n_{o,t-1}$
- Fertility in destination's attractiveness $V_{i,t}$
- Rich interaction between demographic and spatial forces
- $\eta > 1$: migration elasticity

Steady State

Population in all regions grows at common rate g:

$$g = \frac{\sum_{i=1}^{N} n_i L_i}{\sum_{i=1}^{N} L_i}$$

Implications:

- g is endogenous: depends on spatial distribution of fertility
- Policies reallocating population to high-fertility regions raise g
- If $g > 1$: population grows; if $g < 1$: population declines

Model Mechanisms

Local Fertility Elasticities

Proposition: Holding transfers T_i fixed:

$$\varepsilon_{n,w} = \underbrace{-\rho}_{\substack{\text{direct}\\\text{effect}}} \underbrace{-(\gamma-\rho)\tilde{\omega}_i^n}_{\substack{\text{within-nest}\\\text{feedback}}} \underbrace{-(1-\gamma)(1-\omega_i^C)\tilde{\omega}_i^n}_{\substack{\text{cross-nest}\\\text{substitution}}} + \underbrace{\frac{w_i}{w_i+T_i}}_{\substack{\text{income}\\\text{effect}}}$$

$$\varepsilon_{n,r} = \underbrace{\tilde{\omega}_i^h\left[-(\gamma-\rho)-(1-\gamma)(1-\omega_i^C)\right]}_{\text{price effect}}$$

When transfers respond endogenously to rents:

$$\frac{\partial \log n_i}{\partial \log r_i} = \varepsilon_{n,r} + \underbrace{\frac{T_i}{w_i+T_i}\cdot\varepsilon_{T,r}}_{\text{wealth effect }(>0)}$$

Homeowners vs. Renters

Corollary: The relative fertility of homeowners to renters rises with rents:

$$\frac{\partial \log(n_i^O / n_i^R)}{\partial \log r_i} = \frac{\frac{1}{\zeta_i} T_i}{w_i + \frac{1}{\zeta_i} T_i} \varepsilon_{T,r} > 0$$

Intuition:

- Homeowners receive housing revenue transfers
- Higher rents $\Rightarrow$ larger transfers to owners
- Owners' fertility rises relative to renters'
- Consistent with Dettling & Kearney (2014)

Aggregate Fertility Decomposition

Proposition:

$$\hat{\bar{n}} = \underbrace{\sum_i \frac{s_i n_i}{\bar{n}} \hat{n}_i}_{\text{intensive margin}} + \underbrace{\frac{1}{\bar{n}} \mathrm{Cov}_s(n_i, \hat{L}_i)}_{\text{extensive margin}}$$

where s_i is population share of region i

Key insight:

- Extensive margin negative when population flows to low-fertility regions
- Pro-fertility policy in expensive cities may reduce aggregate fertility

Fertility as Stabilizing Force

Proposition: Endogenous fertility dampens population responses to local productivity shocks:

$$\hat{L}_j^{SS}\Big|_{\text{endogenous } n} < \hat{L}_j^{SS}\Big|_{\text{exogenous } n}$$

Mechanism:

- Productivity boom $\Rightarrow$ higher wages, migrants arrive
- Housing costs rise $\Rightarrow$ fertility declines
- Lower fertility partially offsets population growth
- Explains why productive cities haven't grown even faster

Spatial Propagation (1)

Setup: Region j experiences a productivity shock $\hat{A}_j > 0$

Trade channel: Price indices respond across trading partners

$$\hat{P}_i = \sum_k \lambda_{ik} \underbrace{(\hat{w}_k - \hat{A}_k)}_{\text{unit cost}} - \underbrace{\psi \sum_k \lambda_{ik} \hat{L}_k^e}_{\text{agglomeration}}$$

Fertility link: Higher fertility today $\Rightarrow$ larger future $\hat{L}_k^e$ $\Rightarrow$ stronger agglomeration $\Rightarrow$ lower prices for trading partners

Spatial Propagation (2)

Migration channel: Population responds to utility and origin demographics

$$\hat{L}_i = \sum_o \pi_{oi} \left[\underbrace{\eta(\hat{V}_i - \bar{\hat{V}}_o)}_{\text{utility-driven}} + \underbrace{\hat{n}_o + \hat{L}_o}_{\text{origin population}} \right]$$

- Fertility link: Higher $\hat{n}_o \Rightarrow$ more potential migrants $\Rightarrow$ larger inflows to i

Housing channel: Rents respond to demand and population

$$\hat{r}_i = \xi_i \left(\hat{h}_i + \hat{L}_i \right)$$

- Fertility link: Higher fertility $\Rightarrow$ larger $\hat{L}_i \Rightarrow$ higher $\hat{r}_i \Rightarrow$ feedback that *reduces* fertility

Calibration

Data and Parameters

Pre-Determined Parameters	
$\psi = 0.1$	Agglomeration elasticity
$\eta = 1.5$	Migration elasticity (Tombe & Zhu 2019)
$\sigma - 1 = 4$	Trade elasticity
$\theta = 0.5$	Local rental income share
ξ_i	Inverse supply elasticity (Saiz 2010)
Data Sources	
Census 2000	Wages, population, fertility, rents
Commodity Flow Survey	Trade flows (gravity estimation)
Census Migration Tables	County-to-county flows

Preference Parameter Calibration

Identification strategy:

Parameter	Identifying Moment
γ: consumption vs. housing-fertility	corr(n_i, w_i)
ρ: housing vs. fertility	corr(n_i, L_i)
α: consumption share	average fertility
β: housing share	aggregate rent-to-income ratio

Key restriction: Uniform $\chi_i = 0.15$ during calibration

$\Rightarrow$ Forces model to explain variation through wages and rents

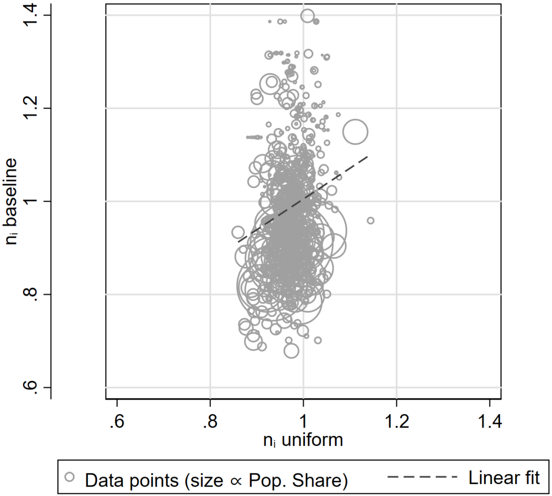
Calibration Results

γ	ρ	α	β
0.91	0.49	0.66	0.60

Key finding: $\rho = 0.49 < 1$

- Strong complementarity between housing and children
- When housing becomes expensive, households cut back on *both*
- Central to understanding fertility-housing relationship

Model Fit: Fertility



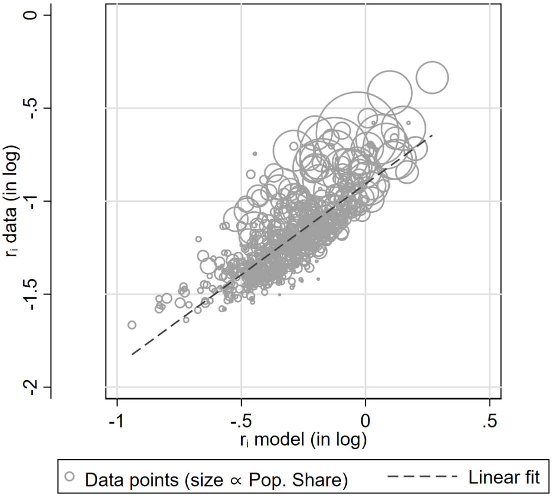
Under uniform χ_i: Slope = 0.66 (s.e. 0.15), $R^2 = 0.03$

Model Fit: Correlations under Uniform χ_i

	Dependent Variable: n_i under uniform χ_i		
	(1)	(2)	(3)
$\log(w_i)$	-0.041^{***}		
	(0.008)		
$\log(L_i)$		-0.004^{***}	
		(0.001)	
$\log(r_i)$			-0.154^{***}
			(0.006)
R^2	0.039	0.038	0.556

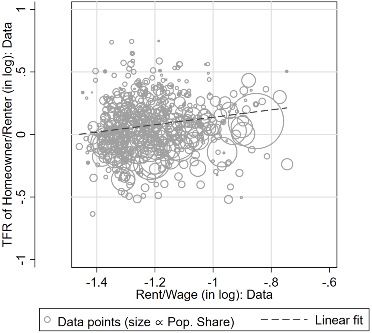
Model reproduces negative correlations from motivating facts

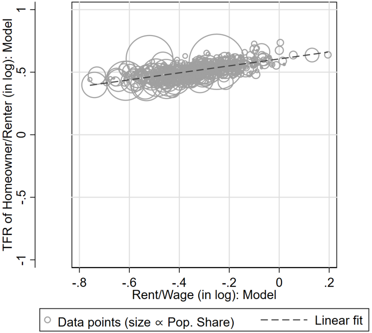
Model Fit: Housing Rents



Slope = 0.98 (s.e. 0.03), $R^2 = 0.61$

Model Fit: Homeowner vs. Renter Fertility





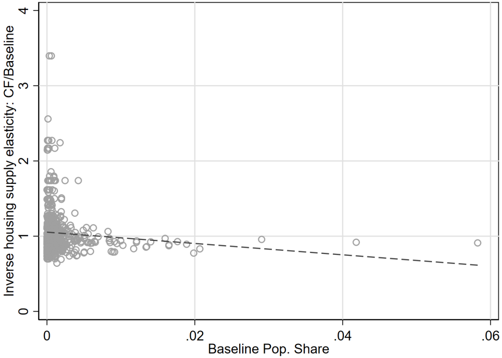
Data: slope = 0.30 (s.e. 0.07); Model: slope = 0.28 (s.e. 0.01)

Counterfactual Analysis

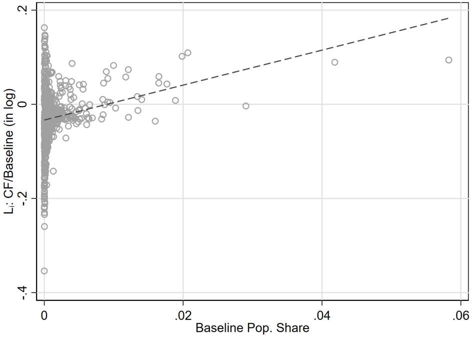
Policy Exercise 1: Housing Supply Reform

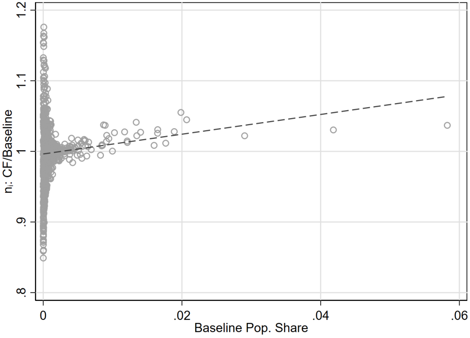
Experiment: Equalize housing supply regulations to median

Effect: Relaxes constraints in large, supply-inelastic cities



Housing Reform: Population and Fertility Effects





- Population increases in large CZs (migration to now-affordable cities)
- Fertility rises in large CZs (housing-fertility complementarity)

Housing Reform: Aggregate Effects

	g	$std(w_i)$	$std(r_i)$	$std(\log L_i)$
Baseline	0.9188	0.1898	0.1294	1.6328
Counterfactual	0.9288	0.1893	0.1633	1.6537

Results:

- Population growth rate rises by 1 percentage point – modest change because large CZs have low fertility baseline
- Wage dispersion falls (spatial convergence)
- Rent dispersion rises (asymmetric policy effects)

Policy Exercise 2: Optimal Spatial Transfers

Consider transfers $t_n \in [0,1]$, $\sum_n t_n = 1$:

$$T_n = \theta \cdot \frac{r_n H_n}{L_n} + t_n \cdot \frac{1-\theta}{L_n} \sum_{k=1}^{N} r_k H_k$$

Two objectives:

1. Per-capita welfare: $\max W^{ave} \equiv \sum_o \frac{L_o}{\sum_{o'} L_{o'}} \left[\sum_k B_k (d_{ok}^{-1} V_k)^\eta\right]^{1/\eta}$
2. Total welfare: $\max W^{tot} \equiv \sum_o L_o \left[\sum_k B_k (d_{ok}^{-1} V_k)^\eta\right]^{1/\eta}$

Optimal Transfers: Outcomes and Tradeoffs

	W^{ave}	W^{tot}	g	$std(w_i)$	$std(r_i)$
Baseline	81.4	55,434	0.919	0.190	0.129
Opt: Per-capita	98.7	62,422	0.853	0.184	0.193
Opt: Total	96.7	63,290	0.883	0.193	0.201

Key tradeoffs:

- Average utilitarian: concentrate workers in few CZs with high productivity, amenity, and child costs, +21% welfare, but fertility falls ($g = 0.85$)
- Total utilitarian: retain workers across more CZs, +19% per-capita welfare, fertility falls less ($g = 0.88$)
- Optimal design depends critically on welfare objective

Conclusion

Model: Spatial equilibrium with endogenous fertility

- Higher wages reduce fertility (substitution $>$ income effect)
- Higher housing costs reduce fertility (housing-children complementarity)
- Local shocks propagate through trade, migration, and housing channels

Empirics: Housing and children are strong complements ($\rho = 0.49$)

Policy implications:

- Housing supply reform: raises both efficiency and fertility (g: 0.92 $\rightarrow$ 0.93)
- Optimal transfers: per-capita objective concentrates workers but sacrifices fertility; total welfare objective balances growth with population

Extensions

- Heterogeneity in worker skills
- Dynamics of human capital formation
- International migration
- Education competition (cf. Arkolakis et al. 2026)

Appendix

Equilibrium Definition

Given fundamentals $(A_i, B_i, d_{oi}, \tau_{in}, \chi_i, \bar{r}_i)$ and parameters, equilibrium consists of:

1. Wages w_i clear labor market
2. Rents r_i clear housing market
3. Housing supply H_i satisfies inverse supply function
4. Price indices P_i from CES aggregation
5. Fertility rates n_i from household optimization
6. Population L_i from migration decisions

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